Serious Game Engines: Analysis and Applications

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***Abstract*—Serious games and their applications in industry, businesses and other 'serious" areas is a fast growing tendency. A success of serious game or serious gamified application's implementation and use in industry significantly depends on quality of external technical gamification platform to be used, or, serious game engine (SGE). The goals of the current SGE Research Group project at Bradley University (Peoria, IL) include an analysis of 100+ serious games and serious gamified applications in industry, development of a comprehensive SGE Comparative Analysis Framework, and classification of SGE and ranking of quality of SGE features, and generation of a set of recommendations on selection and utilization of SGE. This paper presents the main findings and outcomes of the SGE research project.**

Keywords—serious games; game engines; comparative analysis; applications.

I. Serious Games: Definition, Applications in Industry, and Benefits

Advances in computer gaming technology, digital games design, game engines' development, mobile computing, and mobile software engineering stimulated active research on gamification of business processes and activities in "serious" (i.e. not related to entertainment) areas such as simulation of complex physical objects and phenomena, planning, management, marketing, engineering, project management, healthcare, chemistry, and other areas.

We define serious game (SG) as follows: *Serious game is a software system that combines "serious" dimension - a non-entertainment goal and context – and "gaming" dimension, i.e. game design concepts and models, game structure (i.e. game components and links between them), game mechanics (techniques), and game-like graphic user interface* [1].

In accordance with IBM, SGs have capabilities to "…1) solve complex problems collaboratively, 2) make business processes more efficient, 3) achieve predictive modeling and real time visualization, 4) increase ROI from processes, time, and resources, and 5) provide 108% more retention of knowledge, compared to traditional methods" [2]. This motivates companies to design, develop and implement full-scale SG in their business functions and processes. "Business process improvement has been responsible for reducing cost cycle time by as much as 90% while improving quality by more than 60% " [3].

These days many global companies, including Microsoft, IBM, Oracle, Adobe, Cisco, Siemens, SAP, Google, Accenture, American Express, Caterpillar, Ford, etc. already started an implementation of various gamification techniques, gamified applications and entire SG to support their main business functions, processes, and/or activities. In accordance with [4], 53% of surveyed 1,021 technology stakeholders and critics agreed that "…by 2020, there will be significant advances in the adoption and use of gamification".

More than 60 examples of successful use of gamified applications in "serious" areas, serious gamified applications (SGA), and SG by various companies have been reviewed and analyzed [1, 5]; several examples are available in Table I below.

TABLE I. SG and SGA Use in Industry: Examples

Company	Main business activity supported by SG or SGA	Name of SG or SGA
Microsoft	Quality management (improvement of productivity and software quality assurance)	*Language Quality Game, Ribbon Hero*
Microsoft	Quality management (gamification of threat assessment process)	*Elevation of Privilege*
Microsoft	Software development (software development/ coding skills improvement)	*Pex4Fun*
IBM	Communication (corporate social communication and collaboration)	*IBM Connections*
IBM	Planning and management (reduction of cost of internal business)	Document translation
IBM	Management (business process management to drive innovation)	*IBM Innov8*
Cisco	Quality management (improvement of global sales experience)	*The Hunt*
Cisco	Testing (testing of network planning skills)	*myPlanNet*
Cisco	Training and team working (building a team and teamworking)	*The Threshold*
HP	Planning and management (increase of sales)	*Project Everest*
HP	Planning and management (user engagement)	HP Operations Manager
HP	Improvement of conference papers' selection	HP global technical conference
Oracle	Communication (support of communication to about new software features)	*Prune The Product Tree*
PWC	HR management (employee recruitment)	*Multipoly*
Accenture	HR management (employee flexibility)	*Liquid Workforce*
Siemens	Training and development	*Plantville*
SAP	Marketing and branding	*Paul the Octopus*

The outcomes of performed analysis of successful SG and SGA applications in industry clearly show that active use of

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SG and SGA for improvement of business processes amplifies worker's positive psychology, and strengthens and extends the PERMA concepts [6], including:

- positive emotions (experiencing joy, pleasure, fun, safety, etc.),
- engagement (being constantly involved in activities),
- relationships (enjoyable/supportive interactions),
- meaning (creating a purposeful narrative),
- accomplishment (achieving goals, following core values, etc.).

Additionally, SG and SGA placed on corporate web site promote [7]:

- better retention and efficient memorization factors,
- high customization of information,
- better interactivity and maximum implication,
- attractive and non-intrusive advertising,
- better "word of mouth" virality, and
- better dissemination and promotion of web-published information.

Success of SG or SGA implementation and use in industry significantly depends on quality of external technical gamification platform to be used, or, serious game engine (SGE). However, a selection of optimal SGE for SG development is a difficult task [8]; partially, this is because of lack of detailed classification of features of modern SGE.

II. Literature Review

The content of several recent publications [5, 8-10] is related to analysis and classification of features and characteristics of game engines (GE).

The results on GE comparison - Unity, Unreal, Torque, HTML5, Flash, Ogre, Second Life, GameMaker, and XNA - presented in [8] suggest that "serious game developers are primarily using game engines and frameworks that were designed specifically for the creation of leisure or entertainment games. Given the disparity of available resources to serious game developers in comparison to commercial (entertainment) game developers, it is peculiar that they chose the same tools".

A selection framework to identify the ideal GE is based on the technical requirements of the serious game is presented in [9]. The authors compared in details four GE, including CryEngine 3, Unity, Valve's Source Engine, and the Unreal [9], and concluded that "the early-stage selection of the optimal engine for development is crucial".

Several open source GE – JMonkey, Ogre, Unity, Unreal, CryEngine3 - have been reviewed in details in [10]; those GE were primarily analyzed in terms of "modeling of real scenarios in serious games for training. One of the typical uses of serious games (3D serious games) is specialized training in dangerous tasks or when the training is quite expensive. However, typical games use artificial scenarios, created by artists and created according to the restrictions imposed by the game engine used".

Although each of designated and several additional publications used some sort of comparison framework, only certain features of the GE framework were analyzed and compared in those publications. For example, unfortunately, editing tools are only mentioned in those publications; however, a comparison of editing tools is necessary because every SG developer will need and actively use those tools. Only 4-5 SGE are compared in above mentioned publications; however, we believe it is necessary to compare 20-25 SGE in order to provide quality outcomes of comparative analysis. Additionally, the designated publications did not discuss programming experience required to successfully use and work with selected GE.

III. SGE Research Project Goals and Organization

A. SGE Research Project Goals

In 2013-2015, the Serious Game Research Group at Bradley University (Peoria, IL), performed a wide-ranging research project on analysis, design, development and applications of SGE – the SGE Project. Its first project goal was to develop a comprehensive SGE Comparative Analysis Framework (CAF) that ideally should include all features of great importance for SG and SGA developers.

The second SGE Project goal was to identify and analyze SGE of three different groups:

- Group A "excellent and actively used": SGE that a) have a wide set of useful and attractive features for SG developers, and b) have been used for a development of multiple SG; as a result, those SGE are actively used by SG developers.
- Group B "excellent and somewhat used": SGE that also have a wide set of useful and attractive features for SG developers; however, those SGE are not actively used by SG developers' community yet.
- Group C "next generation": SGE that presented various features of next generation SGE; in many cases, those SGE were just released or yet to be released; also, this class of SGE includes game engines which are used by companies alone and may or may not be released to the public.

The third SGE Project goal was to carefully classify identified SGE of three different groups, and, if possible, rank quality of SGE available features from "average" GE developer's point of view.

Finally, the fourth SGE Project goal was, based on obtained research outcomes, to generate a set of recommendations on selection and utilization of SGE for a particular SG design and development project.

B. SGE Evaluation Process

Most of analyzed SGE have been downloaded, installed and carefully tested in research environment, for example Unreal Engine 3, CryEngine 3, Unity, Game Maker Studio, Adobe Flash Professional CC and Torque SGE. The features of the remaining SGE were by analyzing training videos and documentation available at GE-related websites. All SGE features were evaluated by two independent experts in computer gaming; the consolidated average evaluation is presented in Tables IV and V.

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IV. SGE Research Project Outcomes

A. Educational, Simulation and Virtual Reality SG

A brief information about SG and SGA used in industry is presented in Table I. As a part of the next phase of the SGE Project we additionally analyzed 70+ SG in specific areas - 1) education, 2) simulation, and 3) virtual reality areas. Those SG have been designed and developed using various well-known GE; the corresponding statistical data are available in Table II, where VR – Virtual Reality.

TABLE II. SGE Used for SG Development in Selected Areas

Game engine	Educational SG identified	Simulation SG identified	VR SG identified
Unity SIM [13]	12	10	4
Torque [19]	13	2	5
Unreal Engine 3 [11]	1	4	5
Unigine SIM [20]	0	6	4
Neoaxis [16]	3	2	0
CryEngine 3 [12]	0	2	0

B. Proposed Comparative Analysis Framework for SGE

Based on performed analysis of 100+ SGE-based serious games (Tables I and II), the proposed SGE Comparative Analysis Framework is presented in Table III. It is based on creative analysis and integration of framework components used in [5, 8-10] as well as additional proposed evaluation criteria such as a) programming experience required, 2) technical platforms supported, 3) scripting language used, and 4) licenses available.

TABLE III. SGE: Proposed Comparative Analysis Framework

SGE feature	Requirements to SGE features
Graphics	SGE must support both 2D and 3D graphics (these days only a few game engines exist that support both 2D and 3D graphics
Work flow editor	Work flow editor is an important part of SGE - it provides the game developer with no coding experience with an opportunity to create complex actions in the game; it must be built on a concept of object-oriented graphical editor
World (level) editor	World (level) editor, as a crucial SGE component, must help the SG developer to edit or create a map/world using available objects
Character model editor	SGE's model editor should enable the SG developer to create or edit game's 2D or 3D assets/characters
Texture editor	Textures are images which are superimposed on 3D model to give it a realistic feel
Cinematic support	SGE's cinematic support enables the SG developer to create in game movies, where they want a specific incident to happen; thus, game developer becomes a movie director
Physics	Physics – a vital part of SGE; it adds the realistic feel to game play, including lighting, collision detection, solidity of an object, water and cloth physics, weather, etc.
Artificial Intelligence (AI) support	SGE must provide SG developers with complex behavior algorithms (the existing examples are limited to Unreal Engine 3 and CryEngine 3 SGE).
Networking	SGE should provide comprehensive networking support for SG developers to develop advanced and sophisticated distributed, real-time multiplayer games; the key topics in networking include latency, reliability, bandwidth, and security issues
Creation of online game	SGE must be focused on Web-based games that are cross platform systems (these days only a few 3D game engines support this feature)
Programming experience required	SGE must support SG developers with various programming skills; particularly, SG developers with good programming skills should have opportunities to control almost all the aspects of the game engine by themselves. On the other hand, SG developers with low level of programming skills should be supported by SGE by concentrating mostly on content creation and using the work flow editor to add logic in SG
Scripting	SGE should provide SG developer with complete scripting support to write his/her own complex code to optimize SG efficiency in terms of performance and memory usage
Platforms supported	SGE must support easy cross platform deployment of developed SG; however, the SG experienced developers may need to look out for some performance issues of SG on the targeted technical platform
Licensing	SGE licenses can be freeware, open source and commercial; in general, commercial SGE should have significantly grater number of functions fro SG developers

C. Analysis of SGE: Group A

The results of comparative analysis of SGE in Group A are presented in Table IV, group B – in Table V.

The legend used in Tables IV and V is as follows:

5 – "excellent" quality of SGE feature (easy to learn, easy to understand, easy to use, etc.);

4 – "very good" quality of SGE feature ;

3 - "good" quality of SGE feature ;

2 – "satisfactory" quality of SGE feature ;

1 – "low" quality of SGE feature ;

0 – this feature is not present in SGE yet;

low – small number of lines of code or no actual code needs to be written/added, most of the game logic can be added using the work flow editor;

medium – a fair amount of actual coding needs to be done; the SGE graphic user interface will only help with importing of 3D models and audio files;

high - the whole SG needs to be code because the SG editors/tools have very limited capabilities;

free – free version of SGE is available (usually, with limited functionality);

paid – commercial SGE version is available (usually with a maximal set of functions).

TABLE IV. SGE Analyzed: Group A: (part 1)

SGE features	Unreal Engine 3 [11]	Cry Engine 3 [12]	Unity SIM [13]	Game Maker Studio [14]	Adobe Flash Prof. CC [15]
3D graphics	5	5	5	0	3
2D graphics	0	0	4	5	5
Work flow editor	5	5	4	5	0
World editor	5	5	4	4	5

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Character model editor	5	5	5	4	5
Texture editor	5	5	5	4	5
Cinematic support	5	5	5	3	5
Physics support	3	5	4	3	3
AI support	4	4	4	3	2
Networking	4	4	4	5	5
Create online games	0	0	4	4	5
Programming experience needed	Low	Low	Low	Low	Medium
Scripting	Unreal Scrip-ting	Lua	C#, Java Script, Boo	Game Maker Language	Action Script 3.0
Platforms	PC, Mobile, Xbox, Play-Station	PC, Mobile, Xbox, Play-Station	PC, Mobile, Xbox, Play-Station	PC, Mobile, Xbox, Play-Station	PC, Mobile, Web
License	Free, paid	Free, paid	Paid	Free, paid	Paid

SGE ANALYZED : GROUP A (PART 2)

SGE features	Neoaxis Engine [16]	Game-Salad [17]	Cocos 2D [18]	Torque [19]	Unigine SIM [20]
3D graphics	5	0	0	5	5
2D graphics	0	5	5	5	0
Work flow editor	0	5	0	0	0
World editor	5	5	5	5	5
Character model editor	5	5	5	5	5
Texture editor	5	5	5	5	5
Cinematic support	4	3	5	5	5
Physics support	5	4	5	3	4
AI support	3	3	3	4	3
Networking	4	4	4	4	4
Create online games	0	4	0	4	0
Programming experience needed	Medium	Low	Me-dium	Medium	Medium
Scripting	C#	Lua	C++, Java Script	C++	C#, C++
Platforms	PC	PC, Mobile, Web	PC, Mobile	PC, Web	PC, Mobile, Xbox, Play-station
License	Free	Free, paid	Free	Paid	Paid

Subjective evaluations of two independent experts in computer gaming have been used in this research projects; the consolidated average evaluations of each SGE by SG experts are presented in Tables IV and V.

D. *Analysis of SGE: Group B*

The results of comparative analysis of SGE in Group B are presented in Table V.

TABLE V. SGE ANALYZED : GROUP B

SGE features	Quake 4 [21]	Con-struct 2 [22]	Shiva 3D [23]	Delta 3D [24]	Source Engine [25]
3D graphics	4	0	5	5	5
2D graphics	0	5	5	0	0
Work flow editor	0	5	0	0	0
World editor	4	5	5	5	5
Character model editor	4	5	5	5	5
Texture editor	2	5	5	3	5
Cinematic support	3	4	4	4	5
Physics support	4	4	4	4	5
AI support	5	3	3	3	4
Networking	5	5	5	5	4
Create online games	0	4	4	0	0
Programming experience needed	High	Low	Medium	Me-dium	Medium
Scripting	C++	Java Script, HTML	Lua, C++, Objective C	C++	C++
Platforms	PC	PC, Mobile, Web	PC, Xbox, Mobile, Play-station, Web	PC	PC, Xbox
License	Free	Paid	Paid	Free	Paid

E. *Analysis of SGE: Group C*

The results of comparative analysis of SGE in Group C are presented in Table VI.

TABLE VI. SGE ANALYZED : GROUP C

SGE features	CrE 4	UE4 [26]	FB [27]	SD [28]	D2	F	CE6 [29]	ID
3D graphics	x	x	x	x	x	x	x	x
2D graphics		x						
Work flow editor	x	x	x	x		x	x	
World editor	x	x	x	x	x	x	x	x
Character model editor	x	x	x	x	x	x	x	x
Texture editor	x	x	x	x	x	x	x	x
Cinematic support	x	x	x	x	x	x	x	x
Physics support	x	x	x	x	x	x	x	x
AI support	x	x	x	x	x	x	x	x

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Net-working	x	x	x	x	x	x	x	x
Create online games		x						
Program-ming experience	L	L	L	L	L	L	L	H
Scripting	Lua	USc						C++
Platforms	PC, XB, PS	PC, XB, PS, Mob	PC, XB, PS	PC, XB, PS	PC, XB, PS	PC, XB, PS	PC, XB, PS	PC, XB, PS
License	n/a	Sub	n/a	n/a	n/a	n/a	n/a	n/a

The legend used in Table VI is as follows:

CrE4 – CryEngine 4 SGE;
UE4 – Unreal Engine 4 SGE;
FB – FrostBite SGE;
SD – SnowDrop SGE;
D2 – Dunia 2 SGE;
F - Fox SGE;
CE6 – Chrome Engine 6 SGE;
ID – ID tech 5/ID Tech 6 SGE;
X – the feature is available (or will be available soon);
L – no programming skills required;
H – good programming skills required;
Usc – Unreal Scripting programming language;
PC – personal computer-based technical platform;
XB – Xbox technical platform;
PS – PlayStation technical platform;
Mob – mobile technical platform;
Sub – subscription is required;
n/a – information is unavailable yet.

Based on performed research, the SGE of the next generation should provide multiple advanced functions and tools such as

- dynamic day and night;
- photo realistic shading effects;
- procedural destruction;
- GPGPU-based random weather;
- ocean simulator;
- realistic collision and effects, etc.

As a result, it is expected that the next generation SGE will have the following advantageous features:

- efficient use of multi-core CPU [30];
- procedural destruction of objects[31];
- realistic shading effects;
- GPGPU computing support;
- latest Xbox One and PS4 technical platforms.

F. Applications of SGE, SG and SGA in Industry

A quality design of SG or SGA for industrial applications depends on multiple factors, including: 1) existence or absence of corporate software application(s) to be gamified, 2) quality SGE to be used, 3) existence or absence of appropriate SG to implement a particular business function or a process (BFP), and 4) selected set of SG or SGA gamification components to be used in BFP.

The proposed and developed conceptual model of SG and SGA applications in industry is presented in Table VII; it clearly shows a process of construction of various types of SG and SGA for industrial applications.

The following legend is used in Table VII:

BFP - a set of company's business functions and/or processes;

EA - a set of existing software enterprise applications (EA) to support company's BFP;

SG - an external existing fully completed serious game (fro example, IBM's Innov8) that could be used by a company to support its BFPs – either all or part of them;

SGA - a serious gamified application to support company's BFP;

SGE - an external serious game engine that can be used to develop a specialized SG for a company to support its BFPs – either all or part of them;

SGA - a set of enterprise gamified software applications with implemented gamification components such as game mechanics, game design concepts, game-like GUI, etc.; SGA are implemented either directly into current BFP or into existing EA that support BFP.

TABLE VII. CONCEPTUAL MODEL OF SG , SGA AND SGE USE IN INDUSTRY

BFP	EA	SG	SGE	SGA	Out-come	Specific case or details
BFP	EA	-	-	-	EA	Neither SG nor SGA are implemented into EAs to support company's BFP
BFP	-	SG	-	-	SG	Pre-developed external SG is used to support selected company's BFP
BFP	EA	SG	-	-	SG-EA	Pre-developed external SG is implemented directly into EA to support selected company's BFP
BFP	-	-	-	SGA	SGA	Various SGA are implemented to support selected company's BFP
BFP	EA	-	-	SGA	SGA-EA	Various serious gamification components are implemented directly into EA to support selected company's BFP
BFP	EA	SG	SGE	-	SGE-SG	Unique SG are purposely designed and developed using SGE to support selected company's BFP
BFP	EA	-	SGE	SGA	SGE-SGA	Unique SGA are purposely designed and developed using SGE and implemented directly into EA to support selected company's BFP
BFP	EA	SG	SGE	SGA	SGE-	Unique SG and SGA

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					SGA	are purposely designed and developed using SGE and implemented directly into EA to support selected company's BFP

V. Project Findings and Next Steps

A success of SG or SGA implementation and use in industry significantly depends on quality of SGE. The research project completed by the SGE Research Group at Bradley University (Peoria, IL) included 1) analysis of 100+ SG and SGA in industry, 2) development of a comprehensive SGE Comparative Analysis Framework, and 3) classification of SGE and ranking of quality of SGE features, and 4) generation of a set of recommendations on selection and utilization of SGE.

Based on the outcomes of performed research, the up-to-date project findings are as follows:

- SG developers find it difficult to chose an appropriate SGE.
- Unity SIM SGE is recommended for SG developers in the industry. Unity is a powerful SGE with multiple strong features, excellent documentation and tutorials. It supports development of both 2D and 3D games. It is able to create online games – a feature that many SGE do not provide.
- Recommended SGE for 3D game development in academia are Unreal Engine 4 and Unity. The editing tools in these SGE are easy to learn. In case of Unreal Engine 4, students even have access to the game engine source code so that they can implement their own game engine feature. Both SGE have excellent documentation.
- For 2D game development, Adobe Flash Professional SGE is highly recommended for developers who like programming. Learning ActionScript 3.0 is easy as it is very similar to Java programming language.
- For SG developers with basic or no programming experience GameSalad SGE is recommended. GameSalad's intuitive graphic user interface enables rapid prototyping of games with no programming required. GameSalad also includes a discounted educational license.
- Most of SGE are free for educational purposes.
- The next generation SGE are expected to provide SG developers with multiple additional features and tools.

The next steps in this research projects deal with 1) an effective fusion of SGE with smart technology and adding "smart" features to SGE, SG and SGA; 2) use of SG and SGA as "smart" tools to help users to solve "serious" problems based on SGE analytics, sophisticated visualization techniques and smart functions; 3) active integration of SGE, SG and SGA with mobile, social and virtual collaboration systems and technical platforms.

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